WMDR2-core

Package in package 'Model'

WMDR2-core
Version 0.1 Phase 1.0 Proposed
jkl created on 2026-03-06. Last modified 2026-03-11

WMDR2-core diagram

Class diagram in package 'WMDR2-core'

WMDR2core
Notes
1. temporalExtent shall be realized as time:interval
2. temporal<Name> elements other than temporalExtent shall be realized as an array of two arrays, one for times, one for the values of <Name>, e.g., temporalTerritory = [[2000-01-01, 2024-07-13], ["https://codes.wmo.int/wmdr/territory/<ISO1>", "https://codes.wmo.int/wmdr/territory/<ISO2>"]].
3. Following the OGC Features model, geometry = geolocation = coordinates + elevation

version 0.1.4: simplify observation, specifically specification of 1-01

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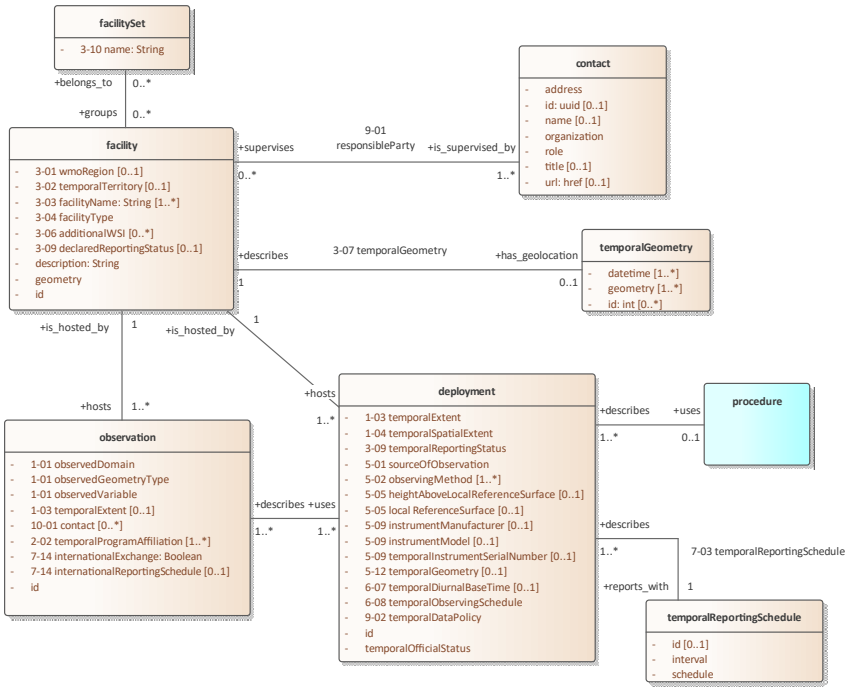

Figure 1: WMDR2-core

contact

Class in package 'WMDR2-core'

impkement according to VCard 4.0 specification [0].

[0] https://datatracker.ietf.org/doc/html/rfc6350#page-23

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ATTRIBUTES

address : Private

ADR: The structured type value consists of a sequence of
address components. The component values MUST be specified in
their corresponding position. The structured type value
corresponds, in sequence, to
the post office box;
the extended address (e.g., apartment or suite number);
the street address;
the locality (e.g., city);
the region (e.g., state or province);
the postal code;
the country name (full name in the language specified in
Section 5.1).

Example:
ADR;TYPE=work:;;Federal Department of Home Affairs FDHA Federal Office of M
eteorology and Climatology MeteoSwiss Measurements and Data\nOperation Cen
ter 1 (OPC 1)\nP.O. Box 257;Zürich-Flughafen;;CH-8058;Switzerland

[Is static True. Containment is Not Specified.]

id : uuid Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

[Is static True. Containment is Not Specified.]

name : Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

Either a full name, or a sequence of name parts. Examples:

Dr. Jörg Klausen (this corresponds to FN in vcard)
Klausen;Jörg;Dr.; (this corresponds to N in vcard)

[Is static True. Containment is Not Specified.]

organization : Private

Example:
ORG:Federal Office of Meteorology and Climatology MeteoSwiss

[Is static True. Containment is Not Specified.]

role : Private

[Is static True. Containment is Not Specified.]

title : Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

ATTRIBUTES

[Is static True. Containment is Not Specified.]

url : href Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

[Is static True. Containment is Not Specified.]

ASSOCIATIONS

Association (direction: Unspecified) 9-01 responsibleParty

Source: Public supervises (Class) facility
Cardinality: [0..*]

Target: Public is_supervised_by (Class) contact
Cardinality: [1..*]

deployment

Class in package 'WMDR2-core'

"deployment" (or "Deployment") refers to the act of placing an instrument (or human observer) at a location to collect measurements or take samples to be analyzed. In many cases, an instrument (or sensor package) can serve different observations.

Deployments can exist in parallel or in series, one after the other.

The intent is to create a new deployment whenever the observingMethod changes over time.

The same deployment may be used if observations are made with the same brand/model of instrument, even if the specific instrument is rotated in and out, for example as part of operational maintenance cycles in a network.

deployment
Version 1.0 Phase 1.0 Proposed
jkl created on 2026-03-06. Last modified 2026-03-18

ATTRIBUTES

1-03 temporalExtent : Private
Alias: records:interval

[Is static True. Containment is Not Specified.]

1-04 temporalSpatialExtent : Private
Alias: records:geometry

Typical spatial geo-referenced volume covered by the observations.

EXAMPLES:
(a) Air temperature at a surface observing site: Sydney Airport, New South Wales, Australia: lat. 33.9465°N, lon. 151.1731°E, alt. 6.0 m above mean sea level (amsl);
(b) The projected area or volume of the cone around a particular weather radar with a maximum range of 370 km (radar reflectivity) and 150 km (Doppler); to be expressed as a geometric shape;
(c) Three-dimensional grid of radar pixels;
(d) For infrared and visible imagery by meteorological satellite (sun-synchronous): global;
(e) For nadir sounding: atmospheric column above ocean;
(f) The Japan Meteorological Agency Advanced Himawari Imager has a 10 min cycle full disk scan (0.5-2.0 km spatial resolution) located above 140.68°E geostationary orbit at an altitude of 35,786 km;
(g) River discharge by gauge: size and geometric shape of a river catchment.

ATTRIBUTES

Q: what is the spatial extent for a)?

[Is static True. Containment is Not Specified.]

3-09 temporalReportingStatus : Private

[Is static True. Containment is Not Specified.]

5-01 sourceOfObservation : Private

[Is static True. Containment is Not Specified.]

5-02 observingMethod : Private
Multiplicity: ([1..*], Allow duplicates: 0, Is ordered: False)

Realized as an array of dictionaries, i.e. key-value pairs:
key: observingMethod id
values: observation id

[Is static True. Containment is Not Specified.]

5-05 heightAboveLocalReferenceSurface : Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

implement as array

[Is static True. Containment is Not Specified.]

5-05 local ReferenceSurface : Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

Linked to heightAboveReferenceSurface

[Is static True. Containment is Not Specified.]

5-09 instrumentManufacturer : Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

Change of instrument requires a new deployment

[Is static True. Containment is Not Specified.]

5-09 instrumentModel : Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

Change of instrument requires a new deployment

[Is static True. Containment is Not Specified.]

5-09 temporalInstrumentSerialNumber : Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

[Is static True. Containment is Not Specified.]

5-12 temporalGeometry : Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

- Geospatial location of instrument/sensor

ATTRIBUTES

[Is static True. Containment is Not Specified.]

6-07 temporalDiurnalBaseTime : Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

indicates the end (hour) of a 24-hour period. Mandatory for 24-hour periodic observations, e.g., cumulative amount of precipitation

[Is static True. Containment is Not Specified.]

6-08 temporalObservingSchedule : Private

complex type: when are observations made? Should be expressed as a temporal coverage, e.g. "from Feb-Oct on working days from 06-18 UTC"

[Is static True. Containment is Not Specified.]

9-02 temporalDataPolicy : Private

remove from WMDR2 and leave for WCMP2?

[Is static True. Containment is Not Specified.]

id : Private

[Is static True. Containment is Not Specified.]

temporalOfficialStatus : Private

[Is static True. Containment is Not Specified.]

ASSOCIATIONS

Association (direction: Unspecified)

Source: Public describes (Class) deployment
Cardinality: [1..*]

Target: Public uses (Class) procedure
Cardinality: [0..1]

ObservingProcedure(s)

Association (direction: Unspecified) 7-03 temporalReportingSchedule

Source: Public describes (Class) deployment
Cardinality: [1..*]

Target: Public reports_with (Class) temporalReportingSchedule
Cardinality: [1]

Association (direction: Unspecified) 5-09 instrument

Source: Public describes (Class) deployment
Cardinality: [0..*]

Target: Public uses (Class) instrument
Cardinality: [0..1]

Association (direction: Unspecified)

Source: Public describes (Class) observation
Cardinality: [1..*]

Target: Public uses (Class) deployment
Cardinality: [1..*]

ASSOCIATIONS

Association (direction: Unspecified)

Source: Public is_hosted_by (Class) facility
Cardinality: [1]

Target: Public hosts (Class) deployment
Cardinality: [1..*]

facility

Class in package 'WMDR2-core'

facility
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ATTRIBUTES

3-01 wmoRegion : Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

mandatory for fixed facilities, optional for mobile facilities.

[Is static True. Containment is Not Specified.]

3-02 temporalTerritory : Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

mandatory for fixed facilities, optional for mobile facilities

[Is static True. Containment is Not Specified.]

3-03 facilityName : String Private
Multiplicity: ([1..*], Allow duplicates: 0, Is ordered: False)

[Is static True. Containment is Not Specified.]

3-04 facilityType : Private

[Is static True. Containment is Not Specified.]

3-06 additionalWSI : Private
Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False)

An array of well-formed WSIs

[Is static True. Containment is Not Specified.]

3-09 declaredReportingStatus : Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

Condition: must be provided if no deployment is specified; otherwise: Optional, inferred as "best case" from deployments: temporalReportingStatus + temporalProgramAffiliation.
Exception: "closed" over-rules deployment, in which case all deployments should be terminated.

[Is static True. Containment is Not Specified.]

description : String Private

ATTRIBUTES

[Is static True. Containment is Not Specified.]

geometry : Private
Alias: 3-07 current geolocation

geometry = geolocation = coordinates + elevation

[Is static True. Containment is Not Specified.]

id : Private
Alias: 3-06 primaryWSI

[Is static True. Containment is Not Specified.]

ASSOCIATIONS

Association (direction: Unspecified)

Source: Public is_hosted_by (Class) facility
Cardinality: [1]

Target: Public hosts (Class) observation
Cardinality: [1..*]

Association (direction: Unspecified) 3-07 temporalGeometry

Source: Public describes (Class) facility
Cardinality: [1]

Target: Public has_geolocation (Class) temporalGeometry
Cardinality: [0..1]

Association (direction: Unspecified) 9-01 responsibleParty

Source: Public supervises (Class) facility
Cardinality: [0..*]

Target: Public is_supervised_by (Class) contact
Cardinality: [1..*]

Association (direction: Unspecified)

Source: Public (Class) facility

Target: Public (InformationItem) temporalGeometry

Association (direction: Unspecified)

Source: Public groups (Class) facility
Cardinality: [0..*]

Target: Public belongs_to (Class) facilitySet
Cardinality: [0..*]

Association (direction: Unspecified)

Source: Public is_hosted_by (Class) facility
Cardinality: [1]

Target: Public hosts (Class) deployment
Cardinality: [1..*]

facilitySet

Class in package 'WMDR2-core'

facilitySet
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ATTRIBUTES

3-10 name : String Private
[Is static True. Containment is Not Specified.]

ASSOCIATIONS

Association (direction: Unspecified)

Source: Public groups (Class) facility
Cardinality: [0..*]

Target: Public belongs_to (Class) facilitySet
Cardinality: [0..*]

instrument

Class in package 'WMDR2-core'

instrument
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ATTRIBUTES

id : Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)
[Is static True. Containment is Not Specified.]

manufacturer : Private

This should come from a registry/catalogue, but should accept any organization.
[Is static True. Containment is Not Specified.]

model : Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

This should come from a registry/catalogue, but can accept "custom-built".
[Is static True. Containment is Not Specified.]

ASSOCIATIONS

Association (direction: Unspecified) 5-09 instrument

Source: Public describes (Class) deployment
Cardinality: [0..*]

Target: Public uses (Class) instrument
Cardinality: [0..1]

observation

Class in package 'WMDR2-core'

observation (or observingCapability) wraps one or several deployments used to generate time series of observations of the same observed variable and the same observedGeometryType. Observed variables can be simple or more complex. Cf. 1-01 observedVariable, 1-01 observedGeometryType for more detail.

observation
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ATTRIBUTES

1-01 observedDomain : Private

cf. http://codes.wmo.int/wmdr/Domain

[Is static True. Containment is Not Specified.]

1-01 observedGeometryType : Private

cf. http://codes.wmo.int/wmdr/Geometry

[Is static True. Containment is Not Specified.]

1-01 observedVariable : Private

The observedVariable is constructed following I-ADOPT ontology approaches, with elements described in a unified vocabulary hosted at codes.wmo.int/wmdr2.
NB: observedVariable will be a LONG list of entries, each record comprised of notation, name, description. The notation is a unique identifier, not necessarily human-readable. The name is potentially human-readable. The description is a fully fledged definition, human-readable and describing all relevant aspects of an observed variable. It is derived from the observedProperty, the object of interest, the matrix and domain, as well as constraints. Matrix and constraints are defined in controlled vocabularies that may be domain-specific.
Examples include:
- observableProperty: state variable (e.g., temperature, pressure, leaf-area-index (LAI), ...), or another property (e.g., thickness, concentration, scattering coefficient, color, ...)
- objectOfInterest: sea ice, CO2, aerosol particles, chlorophyll
- observablePropertyConstraints: a conventional level above ground (e.g., 2m above ground for temperature measurements), a size fraction (e.g., PM2.5), a specific wavelength (e.g., 635 nm), a pressure (range), a temperature (range), a humidity (range), etc., as relevant for the result of the observation. Constraints shall be documented as name-value pairs, where the names shall come from a code list, and the values shall come from a code list depending on the name, or be free text.
- matrix: atmosphere, interstitial air in the root space, air in an ice core, sea surface, soil, dry air, ...

These elements will be supported by code lists from codes.wmo.int/wmdr2.

Applications may want to guide users to select a specific observedVariable by offering groupings, filtering, or tree-like visualizations according to matrix, objectOfInterest, etc.

[Is static True. Containment is Not Specified.]

1-03 temporalExtent : Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

aggregate derived from deployments if deployments are provided; otherwise, specify begin-end

[Is static True. Containment is Not Specified.]

10-01 contact : Private
Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False)

ogc:contact includes "role"

ATTRIBUTES

[Is static True. Containment is Not Specified.]

2-02 temporalProgramAffiliation : Private
Multiplicity: ([1..*], Allow duplicates: 0, Is ordered: False)
[Is static True. Containment is Not Specified.]

7-14 internationalExchange : Boolean Private
[Is static True. Containment is Not Specified.]

7-14 internationalReportingSchedule : Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

complex type: when are observations reported internationally? Should be expressed as a temporal coverage

Conditional: report if internationalExchange==True
[Is static True. Containment is Not Specified.]

id : Private
[Is static True. Containment is Not Specified.]

ASSOCIATIONS

Association (direction: Unspecified)

Source: Public describes (Class) observation
Cardinality: [1..*]

Target: Public uses (Class) deployment
Cardinality: [1..*]

Association (direction: Unspecified)

Source: Public is_hosted_by (Class) facility
Cardinality: [1]

Target: Public hosts (Class) observation
Cardinality: [1..*]

procedure

Class in package 'WMDR2-core'

Extension point to document
- ObservingProcedure
- (Raw)DataProcessing
- ...

procedure
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ASSOCIATIONS

Association (direction: Unspecified)

Source: Public describes (Class) deployment

Target: Public uses (Class) procedure

ASSOCIATIONS

Cardinality: [1..*]

Cardinality: [0..1]

ObservingProcedure(s)

temporalGeometry

Class in package 'WMDR2-core'

temporalGeometry
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ATTRIBUTES

datetime : Private
Multiplicity: ([1..*], Allow duplicates: 0, Is ordered: False)
[Is static True. Containment is Not Specified.]

geometry : Private
Multiplicity: ([1..*], Allow duplicates: 0, Is ordered: False)
[Is static True. Containment is Not Specified.]

id : int Private
Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False)
[Is static True. Containment is Not Specified.]

ASSOCIATIONS

Association (direction: Unspecified) 3-07 temporalGeometry

Source: Public describes (Class) facility
Cardinality: [1]

Target: Public has_geolocation (Class) temporalGeometry
Cardinality: [0..1]

temporalReportingSchedule

Class in package 'WMDR2-core'

temporalReportingSchedule
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ATTRIBUTES

id : Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)
[Is static True. Containment is Not Specified.]

interval : Private

ATTRIBUTES

An array of arrays of begin(inclusive) and end(exclusive) datetime stamps. If the end is not specified, this indicates open-ended. Intervals can overlap to indicate multiple schedules operating in parallel.

[Is static True. Containment is Not Specified.]

schedule : Private

An array of arrays, whereby each nested array corresponds to an interval and describes a schedule. Schedules are specified as follows:

...

[Is static True. Containment is Not Specified.]

ASSOCIATIONS

Association (direction: Unspecified) 7-03 temporalReportingSchedule

Source: Public describes (Class) deployment
Cardinality: [1..*]

Target: Public reports_with (Class) temporalReportingSchedule
Cardinality: [1]

temporalGeometry

InformationItem in package 'WMDR2-core'

temporalGeometry
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ASSOCIATIONS

Association (direction: Unspecified)

Source: Public (Class) facility

Target: Public (InformationItem) temporalGeometry